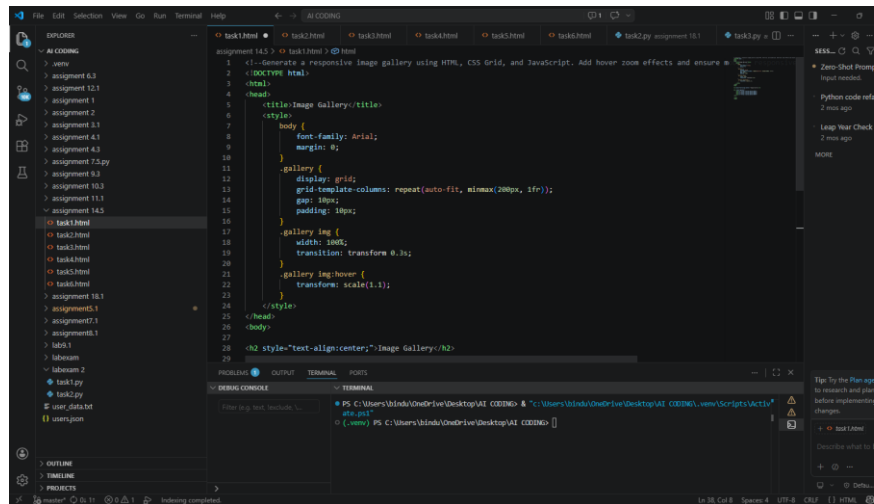


SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
Course Coordinator Name		Dr. Rishabh Mittal	
CourseCode	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
NAME:	M.Anu Chandra	HALLTICKENT NUMBER	2303A54070
Duration	2 Hours	Batche	48
Assignment Number: 14.5			
Q.No.	Question		Expected Time to complete
1	Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation Lab Objectives - Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance. - Apply responsive, accessible, and interactive design principles. - Create practical UI components for real-world web applications. - Use AI to optimize layout, UX, and performance.		Week7 - Friday
	Task 1 – Image Gallery Website Create an image gallery web page. Scenario Create a responsive image gallery using HTML, CSS Grid, and basic styling. The gallery should adjust to different screen sizes and include hover effects. Prompt Used Generate a responsive image gallery using HTML and CSS Grid		

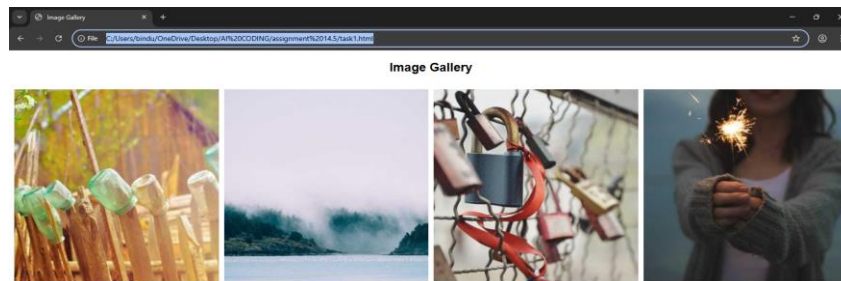
with hover zoom effects and mobile responsiveness.

CODE:



```
1 <!--Generate a responsive image gallery using HTML, CSS Grid, and JavaScript. Add hover zoom effects and ensure w
2 <!DOCTYPE html>
3 <html>
4 <head>
5 <title>Image Gallery</title>
6 <style>
7 {
8   font-family: Arial;
9   margin: 0;
10 }
11 .gallery {
12   display: grid;
13   grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
14   gap: 10px;
15   padding: 10px;
16 }
17 .gallery img {
18   width: 100%;
19   transition: transform 0.3s;
20 }
21 .gallery img:hover {
22   transform: scale(1.1);
23 }
24 </style>
25 </head>
26 <body>
27 <h2 style="text-align:center">Image Gallery</h2>
28
29
```

OUTPUT :



Explanation

This program creates an image gallery using CSS Grid. The images automatically adjust based on screen size. A hover effect is added to make the images zoom when the user points at them.

Task 2 – Profile Card Generator

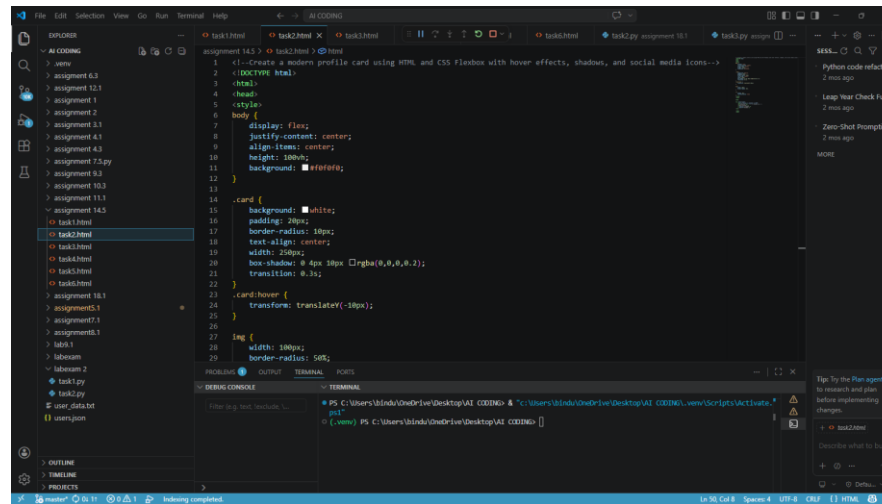
Scenario

Design a profile card using HTML and CSS with hover effects and user details.

Prompt Used

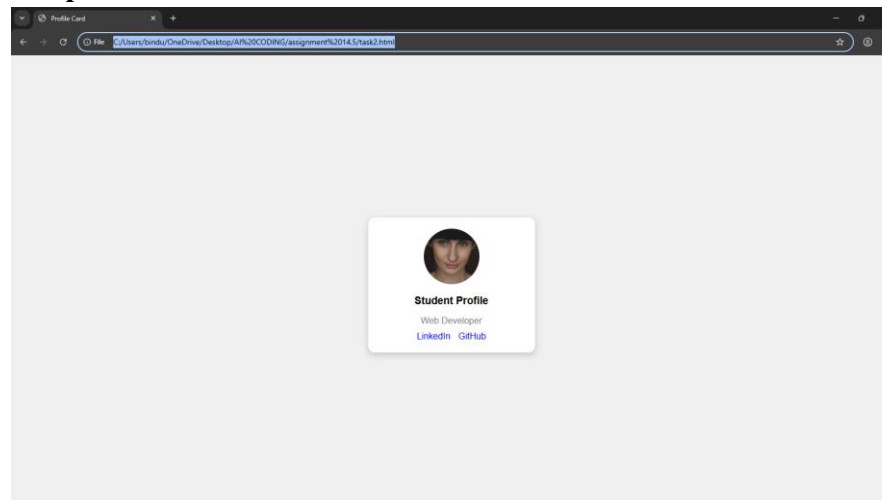
Create a modern profile card using HTML and CSS Flexbox with hover effects and shadow styling.

Code



```
1 <!-- Create a modern profile card using HTML and CSS. Add hover effects, shadows, and social media icons -->
2 <DOCTYPE html>
3 <html>
4 <head>
5 <style>
6   body {
7     display: flex;
8     justify-content: center;
9     align-items: center;
10    height: 100vh;
11    background: #f0f0f0;
12  }
13
14  .card {
15    background: #fff;
16    padding: 20px;
17    border-radius: 10px;
18    text-align: center;
19    width: 200px;
20    box-shadow: 0 4px 8px #000;
21    transition: 0.3s;
22  }
23
24  .card:hover {
25    transform: translateY(-10px);
26  }
27
28  img {
29    width: 100px;
30    border-radius: 50%;
31  }
```

Output:



Explanation:

This program creates a simple profile card. CSS is used to center the card and add shadow effects to make it look attractive.

Task 3 -Develop a dynamic shopping cart system.

Requirements:

Scenario:

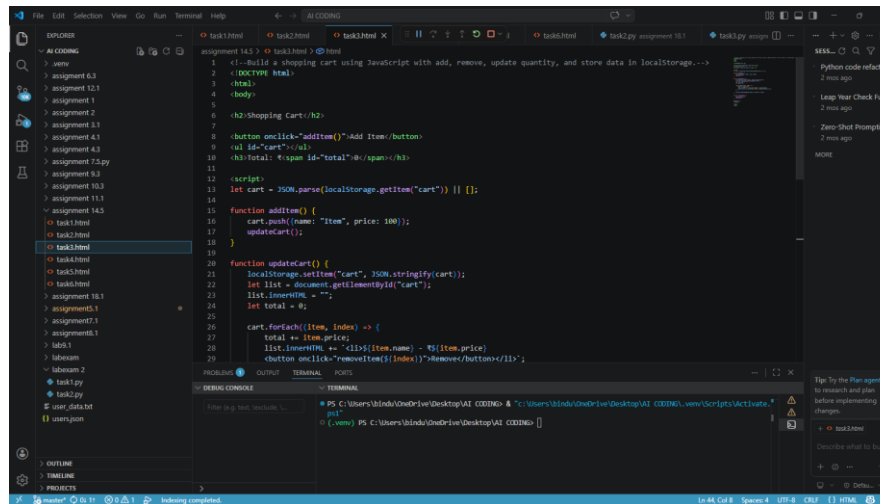
Create a simple shopping cart using JavaScript with add and remove functionality.

Prompt Used:

Build a shopping cart using JavaScript with add and remove item

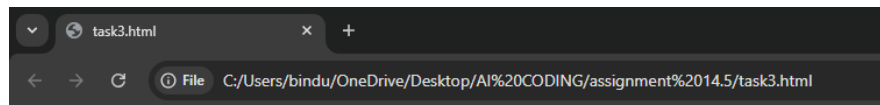
functionality and display total price.

CODE:



```
1 <!--Build a shopping cart using JavaScript with add, remove, update quantity, and store data in localStorage-->
2 <DOCTYPE html>
3 <html>
4 <body>
5
6 <h2>Shopping Cart</h2>
7
8 <button onclick="addItem()">Add Item</button>
9
10 <ul id="cart"></ul>
11 <div>Total: <span id="total"></span></div>
12
13 <script>
14 let cart = JSON.parse(localStorage.getItem("cart")) || [];
15
16 function addItem() {
17   cart.push({name: "Item", price: 100});
18   updateCart();
19 }
20
21 function updateCart() {
22   localStorage.setItem("cart", JSON.stringify(cart));
23   let list = document.getElementById("cart");
24   list.innerHTML = "";
25   let total = 0;
26
27   cart.forEach((item, index) => {
28     total += item.price;
29     list.innerHTML += `<li>${item.name} - ₹${item.price}</li>`;
30     <button onclick="removeItem(${index})">Remove</button>`;
```

OUTPUT:



Shopping Cart

Add Item

Item - ₹100

Remove

Item - ₹100

Remove

Total: ₹200

Explanation

This program uses JavaScript to store items in an array and display them dynamically on the web page.

Task 4 – Role-Based Access Control Application Scenario

Create a system where content changes based on user role (Admin/User).

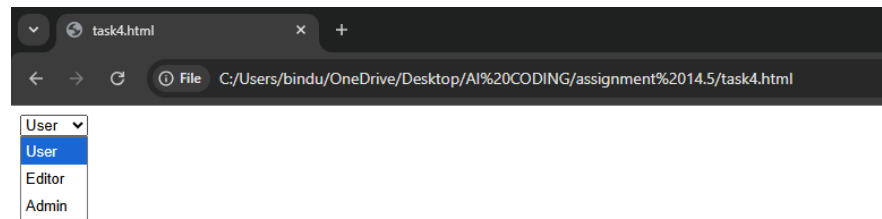
Prompt Used

Create role-based UI using JavaScript with conditional rendering for admin and user roles.

CODE:

```
1 <!-- Create a role-based UI system using JavaScript with Admin, Editor, and user roles and conditional rendering -->
2 <!DOCTYPE html>
3 <html>
4 <body>
5
6 <select onchange="setRole(this.value)">
7 <option value="user">User</option>
8 <option value="editor">Editor</option>
9 <option value="Admin">Admin</option>
10 </select>
11
12 <div id="content"></div>
13
14 <script>
15 function setRole(role) {
16   let content = "";
17
18   if (role === "Admin") {
19     content = "Admin Dashboard";
20   } else if (role === "editor") {
21     content = "Editor Panel";
22   } else {
23     content = "User Page";
24   }
25   document.getElementById("content").innerHTML = content;
26 }
27 </script>
28
29
```

OUTPUT:



Explanation

The program checks the selected role and displays content accordingly using conditional statements.

Task 5 – Multi-Language Website

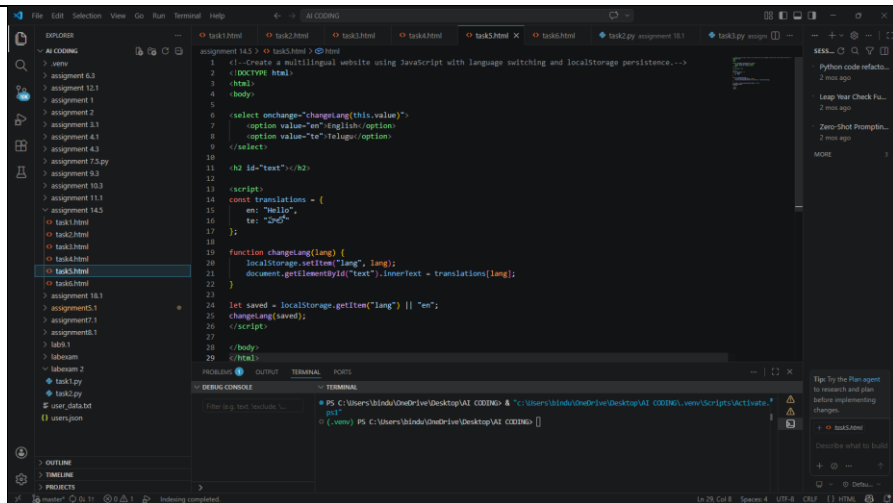
Scenario

Create a website that supports multiple languages.

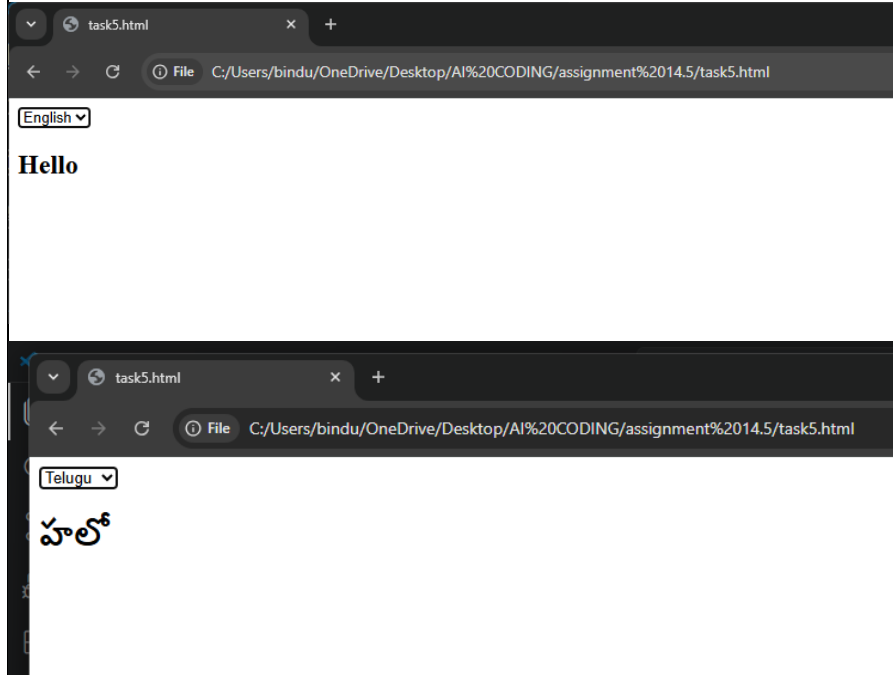
Prompt Used

Create a multilingual website using JavaScript with language switching.

CODE:



OUTPUT:



Explanation

This program switches text based on selected language using JavaScript conditions.

Task 6 – Online Examination System

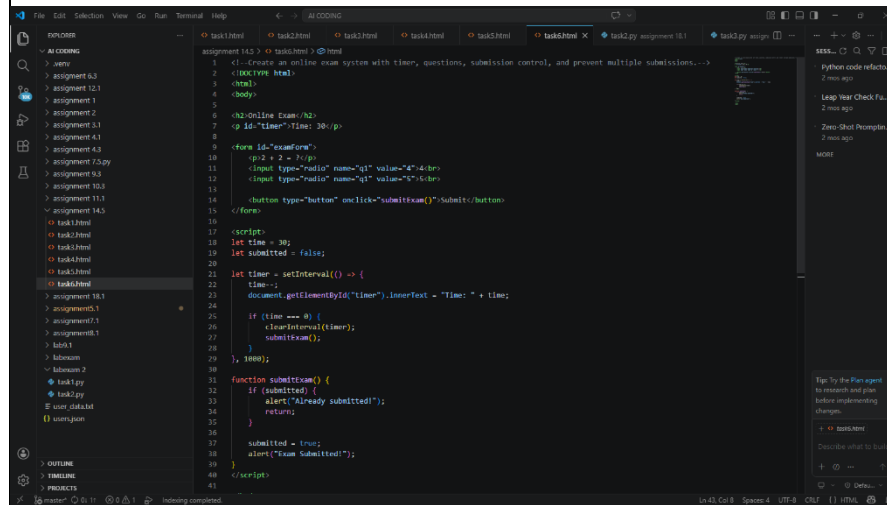
Scenario

Create a simple online exam with a timer.

Prompt Used

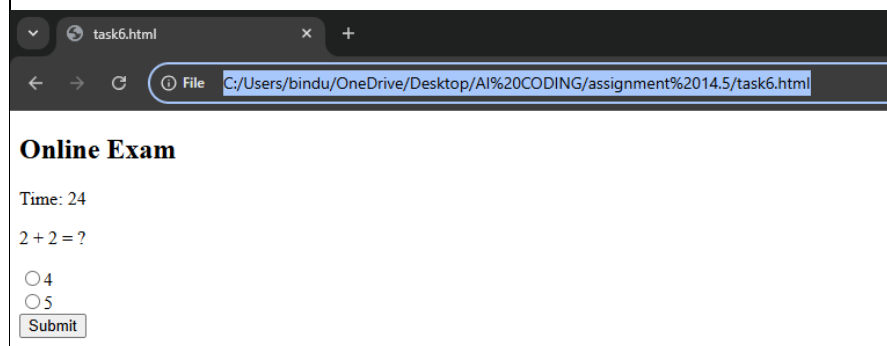
Create an online exam system using JavaScript with timer and submit functionality.

Code



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>
5
6 </Online Exam/>
7 <script>
8
9 <form id="examForm">
10 <p>2 + 2 = ?</p>
11 <input type="radio" name="q1" value="4">4</p>
12 <input type="radio" name="q1" value="5">5</p>
13
14 <button type="button" onclick="submitExam()">Submit</button>
15 </form>
16
17 </script>
18
19 let time = 30;
20 let submitted = false;
21
22 let timer = setInterval(() => {
23   time--;
24   document.getElementById("time").innerText = "Time: " + time;
25 }, 1000);
26
27 if (time === 0) {
28   clearInterval(timer);
29   submitExam();
30 }
31
32 function submitExam() {
33   if (submitted) {
34     alert("Already submitted");
35     return;
36   }
37   submitted = true;
38   alert("Exam Submitted!");
39 }
40
41 </script>
```

Output



task6.html

File C:/Users/bindu/OneDrive/Desktop/AI%20CODING/assignment%2014.5/task6.html

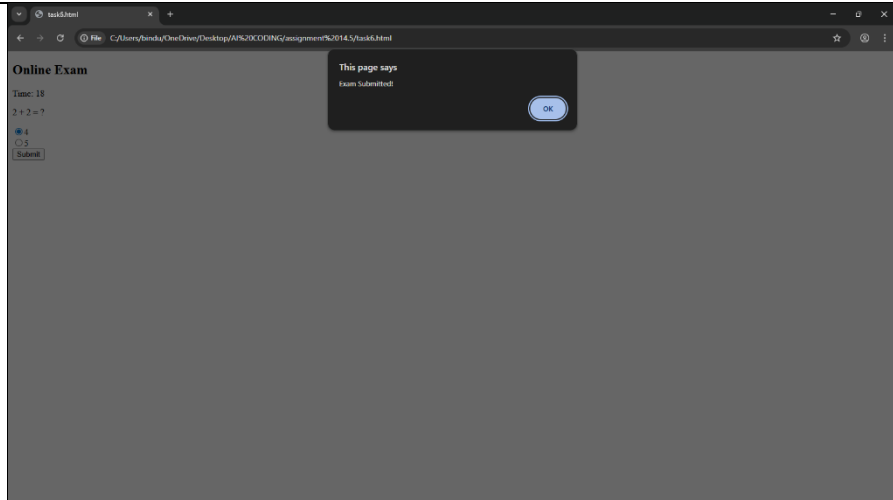
Online Exam

Time: 24

2 + 2 = ?

☐ 4

☐ 5



Explanation

This program uses JavaScript timer to decrease time every second and allows submission.